	Department of Software Engineering	Form: CAF
	School of Electrical Engineering and Computer Science (SEECS)	Date: 31-01-2024
	Corrective Action Request	Course: CS 212

Corrective Action Request for Spring 2024 Semester

Initiated by (Name)	Fahad Ahmed Satti	**Designation / Appointment**	Assistant Professor/ Knowledge Group Head – Programming Core
Nature of Problem:	Rationalization of CLOs and Unified content/programming language, for CS-212 Object Oriented Programming		

Assessment of Problem:

On January 30, 2024, the Programming Core Knowledge Group convened to assess and streamline the Course Learning Outcomes (CLOs) and their alignment with Program Learning Outcomes (PLOs) for CS-212, Object-Oriented Programming. During this session, the Knowledge Group decided to standardize the programming language and course content across the School of Electrical Engineering and Computer Science (SEECS) to ensure a consistent level of expertise among all students. Specifically, CS-212 will emphasize abstract delivery of Object-Oriented Programming (OOP) concepts, subsequently reinforcing these concepts through practical implementation using Java—a pure Object-Oriented Language.

Furthermore, the CLOs underwent a comprehensive review and updating process to accurately align with the appropriate Bloom's Taxonomy (BT) levels, thereby optimizing content delivery. The revised CLOs can be found in Annex A.

Faculty/DLQEC Name: Dr. Fahad Ahmed Satti	**Signature:**

Suggested corrective action(s):

- The centrally aligned outline should be used from Spring 2024.
- The updated CLOs should be adopted for all instances of CS-212.

Name (DLQEC): Mehwish Kiran	**Signature & Date: 01-02-2024**
Name (UG Program Coordinator): Dr. Bilal Ali	**Signature & Date: 01.02.2024**
Name (Head of the Department): Dr. Rafia Mumtaz	**Signature & Date: 01-02-2024**

Annex A

Table 1: Previous CLOs for CS 212 OOP

No.	Course Learning Outcomes	PLO (SE)	PLO (CS)	BT Level
CLO 1	**Understand** core OOP concepts such as classes, objects, encapsulation, inheritance, and polymorphism.	PLO-1	PLO-1	C-2
CLO 2	**Design** solutions to real-world problems by **applying** appropriate object-oriented techniques.	PLO-3	PLO-2	C-6
CLO 3	**Develop** programs using the latest IDEs as per standard practices applicable to the software industry.	PLO-5	PLO-5	P-3
CLO 4	**Demonstrate** commitment to professional ethics by following engineering norms applicable to the software industry.	PLO-8	PLO-8	A-3

Table 2: The proposed CLOs for CS 212 OOP

No.	Course Learning Outcomes	PLO	PLO	BT Level
		(SE)	(CS)	
CLO 1	**Understand** core OOP concepts such as classes, objects, encapsulation, inheritance, and polymorphism.	1 (Engineering Knowledge)	1 (Academic Education)	C-2
CLO 2	**Evaluate** complex programming problems using object-oriented principles analyzing solutions for efficiency and scalability	2 (Problem Analysis)	3 (Problem Analysis)	C-5
CLO 3	**Build** programs using the latest IDEs as per standard practices applicable to the software industry.	5 (Modern Tool Usage)	5 (Modern Tool Usage)	P-3
CLO 4	**Develop** solutions to real-world problems by applying appropriate object-oriented techniques.	3 (Design/Development of Solutions)	4 (Design/Development of Solutions)	C-6
CLO 5	**Demonstrate** commitment to professional ethics by following engineering norms applicable to the software industry.	PLO-8 (Ethics)	PLO-9 (Ethics)	A-3

Faculty OF COMPUTING

FOC DBS Meeting Minutes Held on February 02, 2024

In Attendance

1. Dr. Faisal Shafait (In Chair)	13. Dr. Hashir Moheed Kiyani
2. Dr. Imran Malik	14. Dr. Muhammad Daud Abdullah Asif
3. Dr. Shams Ud Din Qazi	15. Dr. Muhammad Bilal
4. Mr. Maajid Maqbool	16. Dr. Aimal Tariq Rextin
5. Dr. Shah Khalid	17. Dr. Muhammad Moazam Fraz
6. Dr. Syed Imran Ali	18. Dr. Seemab Latif
7. Dr. Mehdi Hussain	19. Dr. Momina Moetesum
8. Dr. Sana Qadir	20. Ms.Naema Asif
9. Dr. Sohail Iqbal	21. Ms. Masabah binte islam
10. Dr. Tahira Lashari	22. Mr. Nadeem Nawaz
11. Dr. Arham Muslim	23. Ms. Iram Tariq Bhatti
12. Mr. Syed Muhammad Ali Musa	24. Ms. Mehwish Kiran

Meeting time: 1500 hrs. – 1600 hrs.

Sr#	Agenda Points / Discussion	Action By
1.	Dr Faisal Shafait welcomed the newly joined faculty member Ms. Mehwish Kiran.	
2.	Dr. Shah Khalid briefed the house about Industrial FYP Intimation on PMS. As per NCEAC guidelines, we are required to have a report on FYPs collaborating with industries. Faculty members are required to use a checkbox during project creation to indicate any industry collaboration.	FoC FMs
3.	Dr. Faisal commenced the meeting by outlining its purpose and addressed the following key points: **a. Uniformity in Course Delivery:** Dr. Faisal emphasized the need for consistent vigor in course delivery across all sections of a course. It was highlighted that the responsibility lies with the KGH to ensure this consistency. To achieve this: ● KGH is tasked with convening a meeting with all Faculty Members (FMs) at the onset of each semester to ensure uniformity in course outlines. The meeting should take place within the first week of the semester commencement. ● KGH must ensure that the depth of course content remains consistent across all sections of the same course. ● KGH must ensure the course outline of UG and PG courses are according to the degree level, PG course content should not be a repetition of the corresponding UG course. **b. Standardization of Examinations:** Dr. Faisal stressed the importance of maintaining a balanced level of difficulty in examinations. It was proposed that: ● Difficulty level and length of the exam papers should be consistent for all courses to ensure fairness. Exam paper vetting by the KGH has to take place one week prior to the commencement of the exam week.	Knowledge Group Head (KGH)

	● FMs teaching different sections of the same course across various degree programs should collaborate in setting exam papers to maintain consistency. ● Each Faculty Member (FM) will be required to submit an undertaking on a Google Form. Within this undertaking, the FM will confirm that the following: i. The question paper they have prepared is distinct from those used in the two preceding offerings of the course. It was determined that the similarity between question papers should not exceed 30%. ii. The exam has been vetted by the KGH. **c. Coherence of Teaching Content:** Dr. Faisal emphasized the significance of coherence in teaching content, particularly for courses that are offered in a particular sequence (e.g. Artificial Intelligence -> Deep Learning; Data Structures & Algorithms -> Design and Analysis of Algorithms. It was decided that: ● KGH should ensure that teaching content is coherent and avoids repetition across sequence courses.	
4.	Dr. Arham highlighted the importance of coherence and standardization in assessment across all courses. Dr. Faisal further elaborated on the topic, emphasizing that ensuring standardization in assessment falls under the purview of the respective Heads of Departments (HoDs). **a. Midterm Review Results Submission:** Dr. Faisal approved the submission of midterm review results by each Faculty Member (FM) in the 10th week of the semester to their respective HoDs. Hence, a Mid-Semester DBS for interim result review will take place. **b. Responsibility of HoDs:** It was agreed that HoDs are responsible for ensuring that both lab and course component assessments are conducted meticulously and meet the required criteria. For each lecture, CH, at least one assignment and two quizzes are expected. **c. Responsibility of FMs and LEs:** Each FMs and LEs are responsible for submitting their assessment to their course HoD in the 10th week of each semester.	KGH and respective course program HoDs FMs and LEs
5.	House showed serious concerns about the lab's component assessment. **a. Lab Engineer Responsibilities:** All labs should be completed within the stipulated time of the lab. LEs should actively engage with the students and not allow any student to leave the lab unless he/she has submitted the lab tasks. They should monitor the student's activities by keeping an eye on their screens to discourage the use of unfair means (e.g. ChatGPT / copying code from peers). b. **Responsibility of Faculty Members (FMs):** The House decided that lab component assessment is the responsibility of the respective FMs. Therefore, it was determined that each lab engineer should get lab evaluation results reviewed by the respective FM. **c. Implementation of Quizzes:**	Lab engineers and Faculty Members

	To enhance assessment effectiveness, it was decided that in the last 15 minutes of each lab session, a 5-10 minute quiz on the same day lab tasks will be conducted on the Learning Management System (LMS) by the lab engineer. It was further stipulated that quiz questions must be designed by the lab engineer and duly approved by the respective FM prior to conduct.	
6.	**Student Advisory:** ● Student Advisors are advised to make sure to keep records of all the correspondence with the students. ● Report the Students who are not attending advisory sessions to ACB.	FoC FMs
7.	**Class Audits:** ● All class auditors are instructed to ensure the quality of class audits.	FoC FMs Class auditors
8.	**Course Outline Approval:** ● LQEC presented the updated course outline for CS212 OOP. ● The House approved the changes. Updated outline is attached in Annex A.	
9.	The department coordinator briefed the house about the new TA hiring policy	



FOC Department Coordinator

February 12, 2024

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Principal SEECS

Annex-A

COURSE OUTLINE

Department:	Faculty of Computing	**Knowledge Group:**	KG-PC	
Programme:	Computer Science (BSCS)	**Class:**	BESE-2K23, BSCS-2K23	
Course code:	CS-212	**Academic Session/Semester:**		Spring 2024
Course name:	Object Oriented Programming	**Pre/co requisite (course name and code, if applicable):**		CS-110 FoCP
Credit hours:	3+1			

Course Synopsis	This course will introduce the Object-Oriented (OO) philosophy to software development, which is a modern and powerful approach. Today, many programming languages support the OO concepts. However, during this course, we'll majorly use the Java programming language. The objective is to make the students understand the benefits of using OO techniques over procedural programming practices, and thereby motivating them to use OO concepts in software development. Further, the students will learn some powerful features of the Java programming language.			
Course Learning Outcomes (CLOs)	1. Understand core OOP concepts such as classes, objects, encapsulation, inheritance, and polymorphism. 2. Evaluate complex programming problems using object-oriented principles analysing solutions for efficiency and scalability. 3. Build programs using the latest IDEs as per standard practices applicable to the software industry. 4. Develop solutions to real-world problems by applying appropriate object-oriented techniques. 5. Demonstrate commitment to professional ethics by following engineering norms applicable to the software industry.			
Course Schedule	Monday 10:00-10:50 Tuesday 10:00-10:50 Thursday 15:00-15:50 Lab (Tuesday 14:00-16:50)			
Course lecturer	**Name**	**Office**	**Contact no.**	**E-mail**
	Mehwish Kiran	A-208		mehwish.kiran@seecs.edu.pk

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	Course Learning Outcomes	PLO (SE)	PLO (CS)	BT Level	Teaching & Learning Methods	Assessment Methods
CLO 1	**Understand** core OOP concepts such as classes, objects, encapsulation, inheritance, and polymorphism.	PLO-1	PLO-1	C-2 (Understand)	Active learning, Cooperative Learning, Blended Learning	Assignment Quiz MSE ESE
CLO 2	**Evaluate** complex programming problems using object-oriented principles analysing solutions for efficiency and scalability.	PLO-2	PLO-3	C-5 (Evaluate)	Active learning, Cooperative Learning, Blended Learning	Assignment Quiz MSE ESE Labs
CLO 3	**Develop** solutions to real-world problems by applying appropriate object-oriented techniques.	PLO-3	PLO-4	C-6 (Create)	Active learning, Cooperative Learning, Blended Learning	MSE ESE Assignments Project Labs

CLO 4	**Build** programs using the latest IDEs as per standard practices applicable to the software industry.	PLO-5	PLO-5	P-3 (Guided Response)	Active learning, Cooperative Learning, Blended Learning	Project Labs
CLO 5	**Demonstrate** commitment to professional ethics by following engineering norms applicable to the software industry.	PLO-8	PLO-9	A-3 (Valuing)	Active learning, Cooperative Learning, Blended Learning	Project Labs

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class or lab activities.
2.	Cooperative learning	Conducted through design project. Students in a team of five will be given a design project that requires software engineering process design solutions. Compliance to the design specifications need to be given in the form of written reports.
3.	Blended learning	Conducted through Learning Management System (LMS) of NUST. All information as well as materials related to teaching and learning activities will be shared with class through this system. Some for formative assessments will be also conducted using this system.

Weekly Schedule:

Week 1	Java Basics • Introduction to Java Programming Language • Compiling & Executing Simple Java Programs • Data types & Operators • Input / Output
Week 2	Flow Control & Composite Datatypes • Decisions • Loops and Iterations
Week 3	Arrays and strings / functions
Week 4 Week 5 Week 6	Classes and Objects • Basic Concepts • Encapsulation & Data Hiding • Access modifiers • Constructors & Destructors • Setters & Getters • Copy Constructor • Default and No-Argument Constructors • Method (Function) Overloading • Static Class Members • this Reference
Week 7 Week 8	Inheritance In Java • Super classes and sub classes • Protected members • Method (Function) Overriding • Constructors in subclasses
Week 9	**Mid-Semester Break**
Week 10 Week 11	Polymorphism • Abstract Classes & Methods • Polymorphic Behaviour

	• Final Methods and Classes
Week 12	Interfaces and Abstract Classes
Week 13	Exception Handling
Week 14	Graphical User Interfaces
Week 15	File Manipulation
Week 16	Project Presentations
Week 17	Revisions
Week 18	**End Semester Break**

Lab Experiments (if applicable):

Lab 1	Java Basics [Data types and Operators]
Lab 2	Flow Control [Loops and Decisions]
Lab 3	Composite data types [Arrays & Strings]
Lab 4	Classes and Objects
Lab 5	Constructors
Lab 6	Composition
Lab 7	Inheritance
Lab 8	Method Overriding
Week 9	**Mid-Semester Break**
Lab 9	Polymorphism
Lab 10	Interfaces and Abstract Classes
Lab 11	Exception Handling
Lab 12	GUI
Lab 13	Event Handling
Lab 14	I/O and File Handling
Lab 15	Open-Ended Lab
Lab 16	Advanced Topics
Week 18	**End Semester Break**

Assessment Methods:

Assessment		Percentage
Theory: 75%		
1	Quizzes (10-15%)	15%
2	Assignments (5-10%)	10%
3	Mid-Term Exam (25-35%)	30%
4	End-Semester Exam (40-50%)	45%
Labs:25%		
1	Labs	70%
2	Project / Final Lab	30%
Total:		100

Learning resources:

Textbook:

1. Java – How to Program (Early Objects), Paul Deitel and Harvey Deitel, 10th Edition, ISBN 978-1- 292-01819-5, Pearson Education, 2015
2. Bruce Eckel, Thinking in Java, Fourth Edition, ISBN-13: 978-0131872486, 25th March, 2006. (available online)

Reference Book:

1. Computing Concepts with Java Essentials, Cay Horstmann, 3rd Edition, ISBN 0-471-24371-x, 2003, John Wiley & Sons.
2. Object Oriented Programming in C++, Robert Lafore, 4th Edition, ISBN-10: 0672323087 | ISBN-13: 978-0672323089, 2001

Grading Policy:

Quiz Policy:

The quizzes will be unannounced / announced and normally last for ten minutes. The question framed is to test the concepts involved in last few lectures. Number of quizzes that will be used for evaluation is at the instructor's discretion.

Project Policy:

Students will be required to develop a project during the course which should be completed towards the end of the semester. They will be graded based on project deliverables and presentation at the end. Students will work in a group/team for projects. A group of 3 students is recommended. At most 4 students are allowed.

Assignment Policy:

In order to develop comprehensive understanding of the subject, assignments will be given. Late assignments will not be accepted / graded. All assignments will count towards the total (No 'best-of' policy). The students are advised to do the assignment themselves. Copying of assignments is highly discouraged and violations will be dealt with severely by referring any occurrences to the disciplinary committee. The questions in the assignment are meant to be challenging to give students confidence and extensive knowledge about the subject matter and enable them to prepare for the exams.

Class participation:

The students are encouraged to participate in class by actively taking part in asking questions from the instructor, sharing his/her thoughts about the topic under discussion, replying to instructor questions, contribute in project presentation and demo. The class participation will be recorded by the instructor and 2% of project marks are assigned to student class participation.

Plagiarism:

SEECS maintains a zero-tolerance policy towards plagiarism. While collaboration in this course is highly encouraged, you must ensure that you do not claim other people's work/ ideas as your own. Plagiarism occurs when the words, ideas, assertions, theories, figures, images, programming codes of others are presented as your own work. You must cite and acknowledge all sources of information in your assignments. Failing to comply with the SEECS plagiarism policy will lead to strict penalties including zero marks in assignments and referral to the academic coordination office for disciplinary action.